



Lab: Draft a Data Retention Policy for an Oil Exploration Company

Estimated time: 30 minutes

Learning objectives

After completing this lab, you will be able to:

- Apply the principles of data retention in enterprise data operations
- Develop a robust data retention policy tailored for an oil exploration company
- Ensure compliance with regulatory, operational, and business requirements

Tools:

You can complete your tasks using any text editor, such as Word, Google Docs, or Notepad.

- If you install Microsoft Word, you can proceed directly with the instructions. However, if you don't have Microsoft Word, you can sign up for a free version of Microsoft 365 online. Follow [these instructions](#) to create your account.
- You can also use Word Online with a Microsoft 365 subscription or access through a free account. If you're new to Word and would like to learn how to get started, you can find helpful resources here: [Getting Started with Word Online](#).
- For this lab, you can use [Lucidchart](#). You only need the free version of LucidChart. If you choose to purchase a paid plan, you will be responsible for any associated charges. If you've not yet signed up for the free account, you can follow the [instructions](#) to get started. You may alternatively use applications like [Microsoft Visio](#).

You will need internet access for research on regulations and standards such as GDPR, CCPA, and industry-specific rules such as EPA or OSHA requirements.

You will also need access to company-specific data governance policies, if available. You will also need a template to create policy documents.

Introduction

In today's data-driven world, managing data assets effectively has become critical for enterprises that deal with huge amounts of data from disparate sources. Saving organizations from unnecessary data hoarding, a data retention policy provides them with a structured approach to managing data by helping them determine how long a certain type of data must be kept for compliance and legal and regulatory purposes, in what format, its access controls, as well as how it must be securely destroyed when it is no longer required.

Exercise: Create a data retention policy for an oil exploration company

In this lab, you will develop a comprehensive data retention policy for an oil exploration company. You will need to perform the following tasks:

1. Identify the types of data managed by the organization
2. Review the applicable regulatory requirements and list the retention timelines for key data categories
3. You will then need to work with key stakeholders to understand the challenges in managing the data lifecycle
4. Next, categorize data types
5. Define the retention periods for each category
6. Create a structured policy document
7. Implement the data retention policy

Scenario

PetroQuest Energy is an oil and gas exploration and production company that collects vast amounts of data from seismic surveys, drilling operations, environmental impact studies, financial transactions, and regulatory compliance documents.

PetroQuest Energy has received a notice from a regulatory authority requesting historical drilling data and environmental impact reports from a specific offshore

project conducted ten years ago. The company is struggling to retrieve the required data due to inconsistent storage practices, missing records, and an outdated document management system.

Simultaneously, a legal dispute has arisen regarding the land rights for one of its past drilling operations. The opposing party claims PetroQuest trespassed beyond its designated area, and are demanding access to geospatial and drilling records from seven years ago. Without a well-defined data retention policy, the company cannot efficiently retrieve the necessary information to defend itself, leading to financial penalties and reputational damage.

Template to use

If you are using Microsoft Word, right-click the link and download the [template](#) document. You can use the template to perform the tasks outlined in the lab.

You can also use the [PDF format](#) for any other word-processing software.

Task 1: Understand the scope and objectives

Step 1: Identify the types of data managed by the organization

Example

Types of data managed by the organization
Drilling reports

Step 2: Define the policy objectives.

Example

Policy objective
Ensure regulatory compliance

Task 2: Review applicable regulations

Step 1: Research and document all the applicable regulatory requirements for environment, finance, health and safety, and global standards

Example

Sectors	Applicable regulatory requirements

Financial	SOX compliance for financial data retention
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Step 2: Summarize retention timelines for key data categories

Task 3: Conduct stakeholder consultations

Step 1: Engage with key stakeholders of the IT and data management, legal and compliance, and business unit teams

Example

Key stakeholders	Purpose of the consultation
IT and data management teams	To understand system capabilities and storage limits

Step 2: Gather feedback on the needs and challenges in managing the data lifecycle

Task 4: Categorize data

Step 1: Classify data into the following categories: critical operational data, compliance data, support data, and archived data

Example

Category	Data
Critical operational data	• Seismic • Well logs • Reservoir data

Step 2: Assign ownership for each data category

Task 5: Define retention periods

Step 1: Establish retention periods for the following categories: regulatory requirements, operational needs, and business records

Example

Category	Retention period

Regulatory requirements	Retain seismic and drilling data for 7-10 years based on industry norms
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Step 2: Specify the format (digital/physical) and storage systems for retained data

Task 6: Outline policy structure

Step 1: Draft a clear and structured policy document, including the purpose and scope, definitions, roles and responsibilities, retention schedule, data archiving and deletion, and policy review and updates

Example

Policy structure	Objective
Retention schedule	Include a table mapping data categories to retention periods

Task 7: Implement the policy

Step 1: Roll out the policy through training sessions for relevant staff

Step 2: Integrate the policy into enterprise systems to configure automated retention rules in data storage systems and set alerts for approaching retention deadlines

Summary

Congratulations on completing the hands-on lab, drafting a data retention policy for an oil exploration company. In this lab, you explored how to develop a comprehensive data retention policy by identifying the types of data and the applicable regulatory requirements. You evaluated the retention periods and finalized the data retention policy for future reference.

It is recommended that you establish mechanisms for periodic audits to ensure compliance with the policy and identify and address any deviations. You can use data analytics to track the efficiency of the retention policy. For example, storage savings, compliance metrics, etc. Compile the finalized policy document, ensuring it includes version control and approval from relevant authorities such as CTO and compliance officer. Store the document in a secure, accessible location for future reference.

Here is the [exemplar](#) document for your reference.

Right-click to open the file. Please remember that the exemplar is just a point of view and not a definitive solution.